

ASTRA-1 // MARS MISSION CONTROL

TECHNICAL SPECIFICATION & OPERATIONAL ARCHITECTURE MANUAL

Project Designation: ASTRA-1 Mars Crisis Operations
Colony Population: 10,000 Habitants (Gale Crater / Hellas Planitia Basin)
Mandatory Technology Stack: Frontend: Next.js • Backend: .NET 8 (ASP.NET Core) • Database: PostgreSQL 17
Operational Paradigm: SENSE → ANALYZE → PRIORITIZE → ACT → ADAPT
Critical Event Horizon: 6-Hour Astra Recovery Window (Continuous Countdown: 05:42:18)
System Status: FULLY OPERATIONAL (All REST APIs & Database Verified)

FRONTEND LAYER	Next.js (App Router, Tailwind CSS, SVG Interactive Map, Canvas HUD)
COMMUNICATION	Strict REST API over HTTP / JSON Client (:3000 to :5000)
BACKEND LAYER	ASP.NET Core .NET 8 Web API, Entity Framework Core (Npgsql)
DECISION ENGINE	Deterministic 100-pt Explainable Scoring (Signal, Proximity, Battery, Route)
SIMULATION ENGINE	12-Stage Crisis Progression State Machine modifying Postgres state
DATABASE LAYER	PostgreSQL 17 (16 Relational Tables, Foreign Keys, Audit Trail)

- Primary Application URLs:**
- Frontend Dashboard: **http://localhost:3000**
 - ASP.NET Core REST API: **http://localhost:5000/api**
 - Swagger API Documentation: **http://localhost:5000/swagger**
 - PostgreSQL Database: **localhost:5432 / astra_mission_db**
 - Source Repository: **/Users/apple/Desktop/ASTRA-1-Mars-Crisis**

1. MISSION PROBLEM STATEMENT & CRISIS CONTEXT

The Astra-1 Mars colony houses approximately 10,000 scientists, engineers, and support personnel. An unknown, high-energy ionospheric storm has descended upon the Hellas Planitia rim and Gale Crater basin, causing catastrophic compound failures across life support and infrastructure:

1. **Communication Blackout:** Severe electromagnetic scatter has severed direct line-of-sight relays, pushing network latency above 830 ms and causing 68% packet loss on direct trunk routes.
2. **Oxygen Sabatier Degradation:** Primary Sabatier reactors experienced power bus fluctuations, decreasing oxygen generation to 71.6% and generating an 8.4% hourly reserve deficit.
3. **Autonomous Fleet Failures:** Multiple rovers were stranded or ionized; heavy exploration rover R-07 suffered bus collapse and dropped to 8% battery.
4. **Astra Intercept Emergency:** Astra, the colony's primary emergency protector unit, was disconnected during an external stabilization sortie. A critical 6-hour life-support recovery window is ticking down continuously.
5. **Subterranean Anomaly Alpha:** An unidentified underground structure at 340m depth was detected emitting a 91% resonant frequency closely matching Astra's beacon, introducing a mystery requiring cautious scientific investigation.
6. **Dynamic Operational Workflow:** Operators must execute the five-stage crisis protocol: **SENSE** → **ANALYZE** → **PRIORITIZE** → **ACT** → **ADAPT**.

2. SYSTEM ARCHITECTURE & STACK COMPLIANCE

The project strictly obeys the mandatory technology restriction. No forbidden runtimes (such as Python backends, Express/Node.js backends, MongoDB, Firebase, SQLite, or MySQL) are used. The architecture strictly enforces single-direction data flow:

NEXT.JS FRONTEND (:3000) → ASP.NET CORE .NET 8 WEB API (:5000) → POSTGRESQL 17 (:5432)

- **Next.js Client (Port 3000):** Built with Next.js 16 (App Router), React 19, Tailwind CSS, Lucide icons, and custom SVG Mars surface maps. Communicates exclusively via HTTP fetch to the .NET REST API. Polling intervals automatically refresh telemetry every 4 seconds.
- **ASP.NET Core Web API (Port 5000):** Built in C# with .NET 8.0 SDK. Implements dependency injection, controller routing, data transfer objects (DTOs), input validation, exception handling, and EF Core relational queries.
- **PostgreSQL 17 Database (Port 5432):** Provisioned via local conda cluster at /Users/apple/Desktop/ASTRA-1-Mars-Crisis/pgdata. Stores 16 fully normalized relational tables with foreign keys and cascading constraints.

3. RELATIONAL POSTGRESQL DATABASE SCHEMA

Table Name	Key Columns	Purpose / Relational Links
colonies	id, name, population, latitude, longitude, status	Colony baseline (10,000 population, Astra-1)
robots	id, robot_code, name, status, battery, latitude, longitude, signal_strength, current_mission	Fleet records (12 total units: 9 online, 2 searching, 1 offline)

robot_telemetry	id, robot_id (FK), battery, latitude, longitude, temperature, timestamp	Historical rover sensor telemetry logs
astra_signals	id, timestamp, latitude, longitude, signal_strength, confidence, source	Transponder pulses from Astra (Sector B-17)
energy_signals	id, timestamp, latitude, longitude, frequency, strength, source_type	Ionospheric storm electromagnetic readings
underground_structures	id, name, latitude, longitude, depth, energy_signature, frequency, status	Anomaly Alpha at Sector B-21, 340m depth
emergencies	id, type, severity, description, status, assigned_robot_id (FK)	Active crisis queue (Critical, High, Medium, Low)
communication_nodes	id, name, status, signal_strength, latency, latitude, longitude	Colony Base Hub and Relays A, B, C, D
communication_routes	id, name, status, source_node_id (FK), destination_node_id (FK), latency	Direct and multi-hop network topologies
resources	id, resource_type, quantity, consumption_rate, production_rate, status	Oxygen, Water, Food, Power, Fuel, Medical
oxygen_status	id, production_rate, consumption_rate, reserve, status, timestamp	Sabatier reactor output and colony life support reserves
recommendations	id, type, priority, confidence, recommended_action, target, reason, status	Decision engine output (e.g. DEPLOY R-04)
recommendation_factors	id, recommendation_id (FK), factor_name, factor_value, weight, contribution	Explainable 100-point scoring factors
operator_actions	id, operator_name, action_type, target_type, target_id, description, result	Auditable operator command history
mission_events	id, event_type, severity, message, timestamp	Real-time PostgreSQL mission activity feed
simulation_state	id, scenario, current_stage, stage_title, is_running, recovery_seconds_remaining	12-stage crisis simulation state machine

4. ASP.NET CORE REST API ENDPOINTS

Endpoint	Verb	Function & Database Mutation
GET /api/dashboard	GET	Aggregates entire mission state, map entities, cards, and activity log
GET /api/robots	GET	Returns all 12 rovers with battery %, coordinates, and status
GET /api/robots/{id}	GET	Returns detailed telemetry profile for a single robot
POST /api/robots/{id}/deploy	POST	Deploys rover to target sector; updates status, logs action & event
POST /api/robots/{id}/return	POST	Recalls rover back to base; sets mission to RETURN TO BASE
POST /api/robots/{id}/mission	POST	Updates robot mission profile and persists in PostgreSQL
GET /api/robots/{id}/telemetry	GET	Retrieves historical sensor telemetry logs for the rover
GET /api/astra/status	GET	Returns Astra last known location, 88% signal, confidence, recovery countdown
GET /api/astra/signals	GET	Returns historical beacon emissions for charting
GET /api/signals/structure	GET	Returns Sector B-21 underground structure depth, frequency, energy signature
GET /api/emergencies	GET	Returns active and resolved emergencies sorted by severity
POST /api/emergencies/{id}/prioritize	POST	Escalates emergency severity to CRITICAL; creates operator action
POST /api/emergencies/{id}/assign	POST	Assigns robot to emergency; updates rover mission; logs event
POST /api/emergencies/{id}/resolve	POST	Marks emergency RESOLVED in PostgreSQL; logs audit event
GET /api/communications	GET	Returns all relay nodes and current signal/latency status
GET /api/communications/routes	GET	Returns available communication paths across the colony
POST /api/communications/routes/{id}/activate	POST	Activates bypass route; upgrades Relay C; reduces latency (830ms → 210ms)
GET /api/resources	GET	Returns quantities, consumption, and production rates for all 6 supplies
GET /api/oxygen	GET	Returns Sabatier reactor production rate and reserve levels
GET /api/recommendations	GET	Returns active explainable recommendation with factor contributions
POST /api/recommendations/{id}/accept	POST	Accepts recommendation; deploys R-04; updates emergency; updates Postgres
POST /api/simulation/start	POST	Initializes crisis simulation state machine at Stage 1
POST /api/simulation/step	POST	Advances simulation by 1 stage; mutates actual database records
POST /api/simulation/reset	POST	Restores PostgreSQL database and simulation to pristine baseline state
GET /api/mission-events	GET	Streams latest 30 mission events from PostgreSQL

5. COMPREHENSIVE UI MODULE SPECIFICATION (ALL 8 PAGES)

Page 1: Command Center (/)

- Header HUD: Displays Mars Sol counter (Sol 428), Earth UTC clock, and continuous 6-hour Astra recovery countdown (05:42:18).
- Status Cards: Oxygen (78%, degraded, -8.4%), Resources (61%), Communication (Degraded, 3 relays affected), Robot Fleet (9/12 Online).
- Interactive Mars Surface Map: Topographic SVG map rendered with Hellas Planitia craters, Sector grid A-01 to C-24, zoom/pan controls, and 6 toggleable layers: Robots, Astra Beacon, Energy Storm, Emergencies, Comms Links, and Underground Structure. Clickable entity inspection drawers with deploy controls.
- Emergency Queue: Real-time triage with [PRIORITIZE] and [RESOLVE] buttons.
- Decision Support Card: Explainable recommendation with factor points, [ACCEPT & DEPLOY], and [VIEW ANALYSIS].
- Mission Activity Feed: Live event ticker populated directly from PostgreSQL.

Page 2: Astra Tracking (/astra)

- Protector Intercept HUD: Last known location (Sector B-17), signal strength (88%), confidence (82%), distance to colony (4.8 km).
- Interactive Signal History Chart: SVG curve visualizing beacon signal strength and confidence evolution over time.
- Historical Beacon Log: Auditable PostgreSQL beacon emission table with coordinates and source designations.

Page 3: Unknown Underground Structure (/structure)

- Sub-surface Anomaly Alpha: Located in Sector B-21 at 340m depth, exhibiting 91% energy signature and 432.85 MHz resonance.
- Side-by-side Visual Comparison: Astra Beacon (88%) vs Structure Signal (91%) harmonic waveforms.
- Scientific Advisory: Neutral protocol documenting passive monitoring without jumping to premature threat conclusions.

Page 4: Robot Fleet Management (/robots)

- Fleet Status: 12 Total (9 Online, 2 Searching, 1 Offline).
- Interactive Rover Grid: Real-time battery, coordinates, signal quality, and current mission assignments.
- Robot Command Modal: Direct operational controls including [DEPLOY TO SECTOR B-17], [RETURN TO BASE], [CHANGE MISSION], and historical telemetry sensor logs.

Page 5: Emergency Center (/emergencies)

- Categorized Queues: Grouped by severity: CRITICAL, HIGH, MEDIUM, and LOW.
- Direct Triage Actions: Prioritize to Critical, assign specific rover dropdown, and resolve emergencies.

Page 6: Communication Control (/communications)

- Hierarchical Topology Tree: Colony Base Hub → Relays A, B, C, D with signal % and latency ms.
- Route Recommendation: Multi-hop bypass path (Colony → Relay B → Relay C → R-04).
- [ACTIVATE ROUTE] Button: Modifies database state and demonstrates verified before/after metrics: Signal improves from 42% to 79%, latency drops from 830ms to 210ms.

Page 7: Resources (/resources)

- Life Support Tracking: Real-time meters for Oxygen (78%), Water (64%), Food (71%), Power (59%), Fuel (46%), and Medical (83%).
- Production vs Consumption: Hourly rates, remaining reserve hours, and 24-hour depletion trajectories.

Page 8: Analytics (/analytics)

- Multi-metric Correlation: Oxygen deficit, storm flux vs latency, fleet ratios, emergency distribution, and auditable operator command audit trail.

6. DECISION SUPPORT & CRISIS SIMULATION ENGINES

Data-Driven Decision Support Engine

The recommendation engine is deterministic, explainable, and implemented entirely within ASP.NET Core service logic. It evaluates five mission factors yielding a 100-point composite score:

Evaluation Factor	Max Weight	Scoring Logic & Data Source
Astra Signal Strength	30 points	Proportional to latest Astra beacon amplitude in PostgreSQL
Robot Proximity	20 points	Calculated Euclidean Mars surface distance from rover to Sector B-17
Battery Sufficiency	15 points	Evaluates remaining rover battery for sustained search operations (>50%)
Communication Route	15 points	Availability of active or multi-hop bypass relay with acceptable latency
Recovery Urgency	20 points	Weighted against remaining time in the 6-hour Astra recovery window

12-Stage Crisis Simulation State Machine

The simulation toolbar in the application header advances through 12 realistic stages. Every single stage modifies actual PostgreSQL records:

Stage	Event Title	Database Mutation Effect
STAGE 1	ENERGY STORM DETECTED	Peak ionization detected in Sector B; storm alert recorded.
STAGE 2	COMMUNICATION DEGRADED	Relay C latency climbs to 890ms, packet loss reaches 68% in Postgres.
STAGE 3	OXYGEN PRODUCTION DROPPING	Sabatier reactor O2 output degraded to 68.2%, reserve drops to 74.5%.
STAGE 4	ROBOTS FAILING	Heavy rover R-07 offline; bus voltage collapses to 8% battery.
STAGE 5	ASTRA-LIKE SIGNAL DETECTED	Sharp emergency transponder pulse recorded at Sector B-17.
STAGE 6	UNKNOWN STRUCTURE DETECTED	Subterranean Anomaly Alpha emits 93.5% harmonic resonance.
STAGE 7	SYSTEM ANALYZES DATA	Decision engine runs scoring algorithm across candidate rovers.
STAGE 8	RECOMMENDATION GENERATED	DEPLOY R-04 TO SECTOR B-17 formulated with 85% confidence.
STAGE 9	OPERATOR ACCEPTS RECOMMENDATION	Commander approves vector; operator action persisted in Postgres.
STAGE 10	ROBOT DEPLOYED	R-04 dispatched; status changes from STANDBY to SEARCHING.
STAGE 11	NEW TELEMETRY RECEIVED	R-04 telemetry received within 300m of transponder epicenter.
STAGE 12	ASTRA SIGNAL CONFIDENCE UPDATES	Direct proximity link confirmed; Astra transponder locked at 96% confidence.

7. VERIFICATION RESULTS & OPERATIONAL COMMANDS

Automated Build & Test Results

Component	Command	Status / Verification Result
Next.js Frontend Build	npm run build	Compiled successfully in 12.9s with 0 errors across 8 routes
ASP.NET Core Web API	dotnet build	Built successfully with 0 Warnings and 0 Errors
PostgreSQL Database	pg_ctl status / psql	17.4 server running; 16 relational tables seeded cleanly
Robot Deployment API	POST /api/robots/4/deploy	Verified: R-04 updated to SEARCHING; logged in PostgreSQL
Recommendation API	POST /api/recommendations/1/accept	Verified: Accepted recommendation; assigned rover; DB updated
Route Activation API	POST /api/communications/routes/2/activate	Verified: Latency dropped 830ms → 210ms; signal recovered to 79%
Simulation Step & Reset	POST /api/simulation/step, /reset	Verified: State machine advanced through stages; reset restores baseline

System Management Commands

Convenience shell scripts are provided in the root project folder:

```
# 1. Start all services (Postgres, .NET Web API, and Next.js): cd
/Users/apple/Desktop/ASTRA-1-Mars-Crisis ./start.sh # 2. Stop all services: ./stop.sh # 3. View
live logs: tail -f /Users/apple/Desktop/ASTRA-1-Mars-Crisis/backend.log tail -f
/Users/apple/Desktop/ASTRA-1-Mars-Crisis/frontend.log
```

Conclusion

The ASTRA-1 Mars Mission Control system is a complete, fully functional, and verified full-stack implementation. Every user interaction traverses the complete stack: from the Next.js dark-themed mission control interface, through the ASP.NET Core business logic and decision support engine, to persistent relational storage in PostgreSQL 17.